Financial Economics – Lesson 1: Financial Markets

Giulio Bottazzi and Daniele Giachini

PhD course, Scuola Superiore Sant'Anna

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Outline

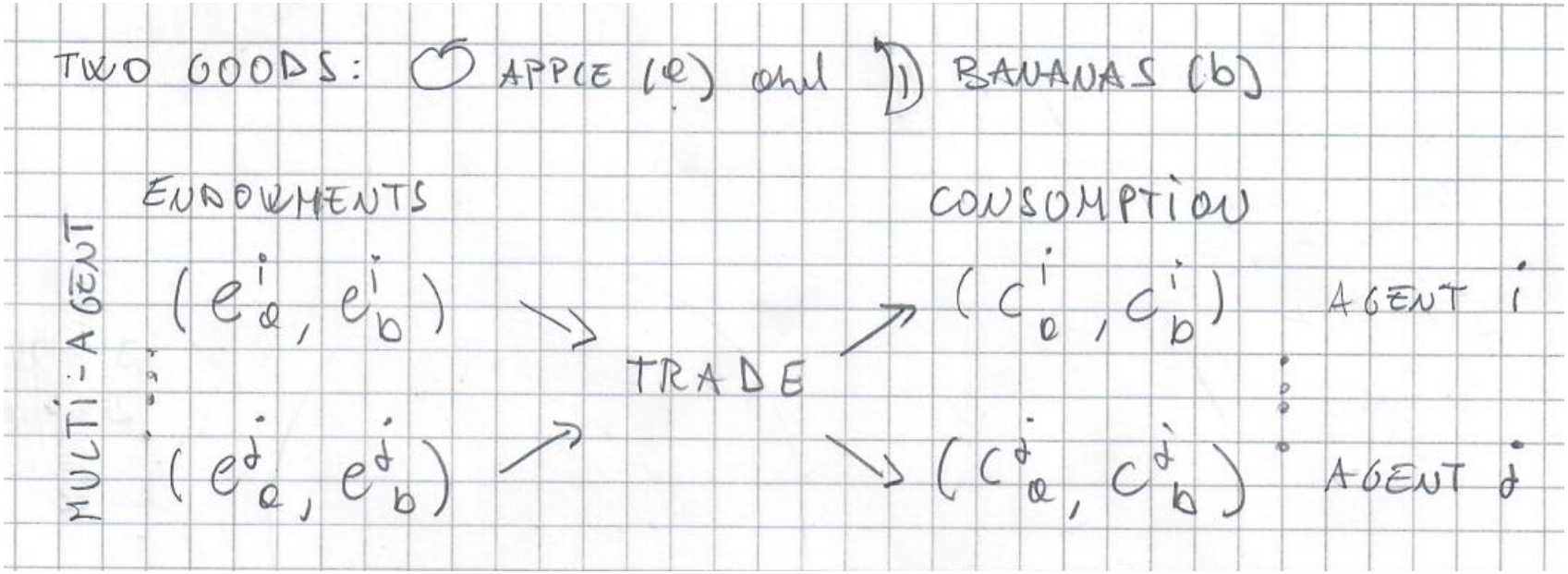
Introduction

- A review of fixed income with drawings
- What is Financial Economics?

Two-Date model

- Linear Pricing
- Arbitrage and Positive Pricing
- No-arbitrage Valuation
- Risk neutral probabilities

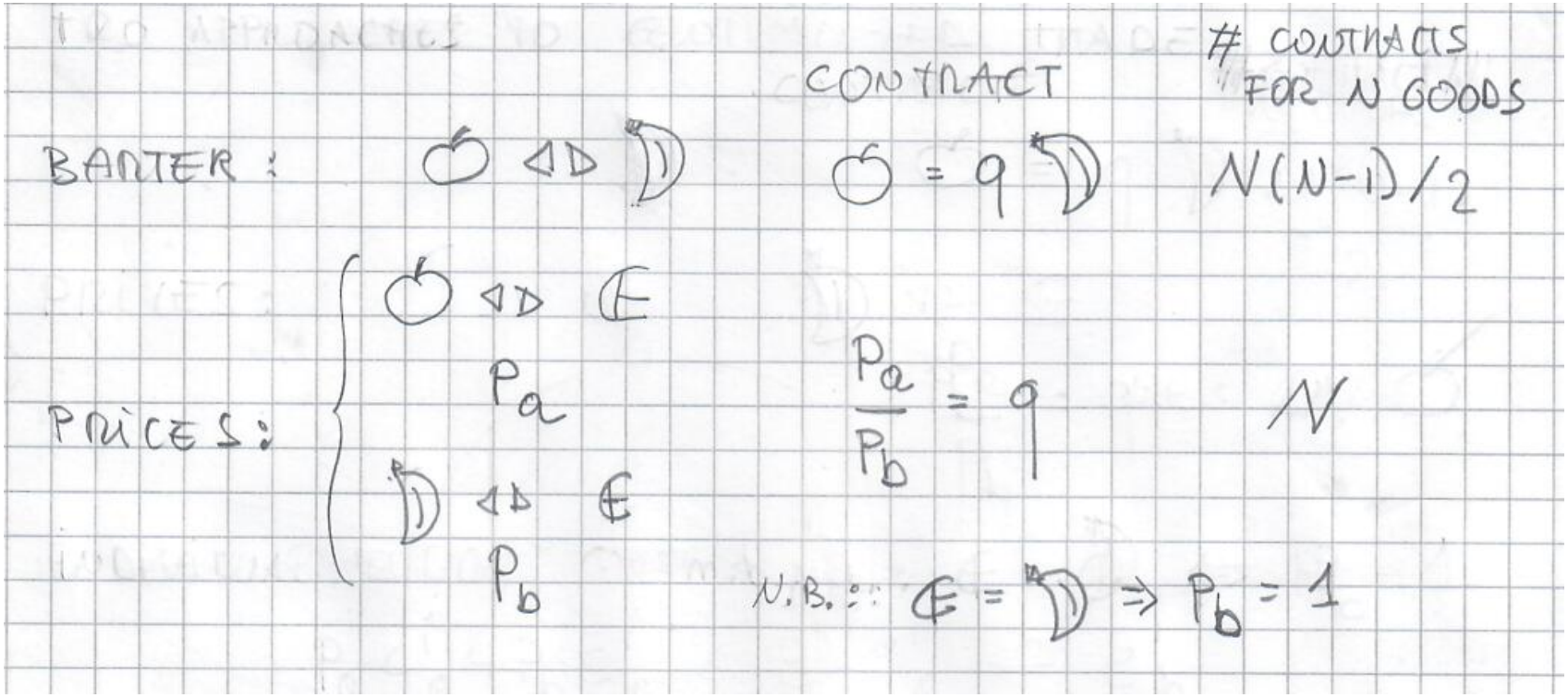
Trade and market clearing



Consumption market clearing conditions

$$\sum_i c^i_a \leq \sum_i e^i_a \,, \quad \sum_i c^i_b \leq \sum_i e^i_b \,.$$

Centralized setting

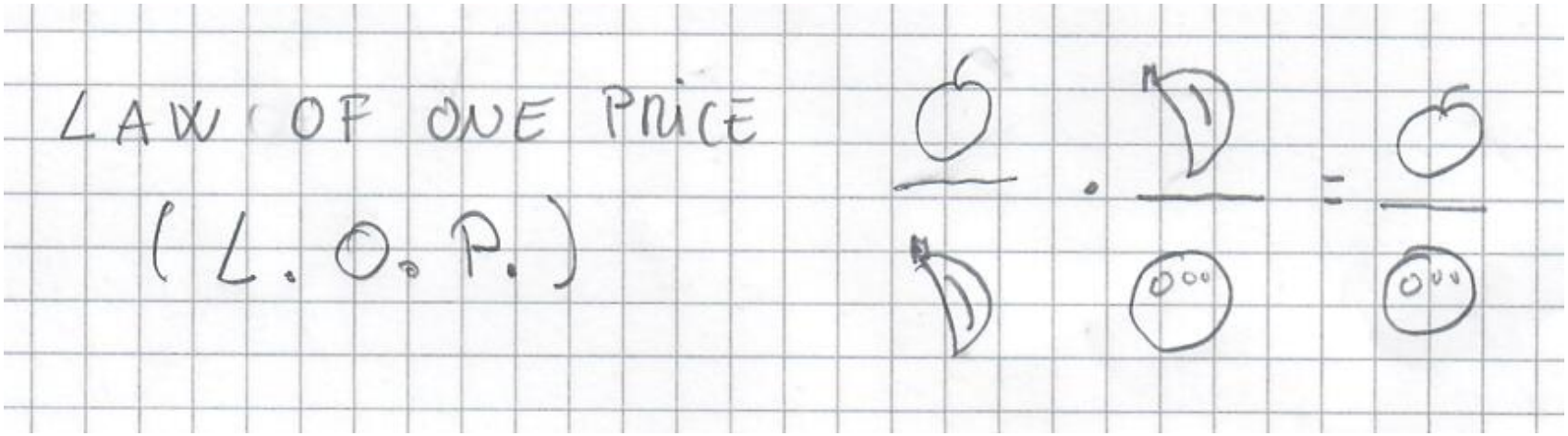


With prices, individual budget constraint, for each agent i:

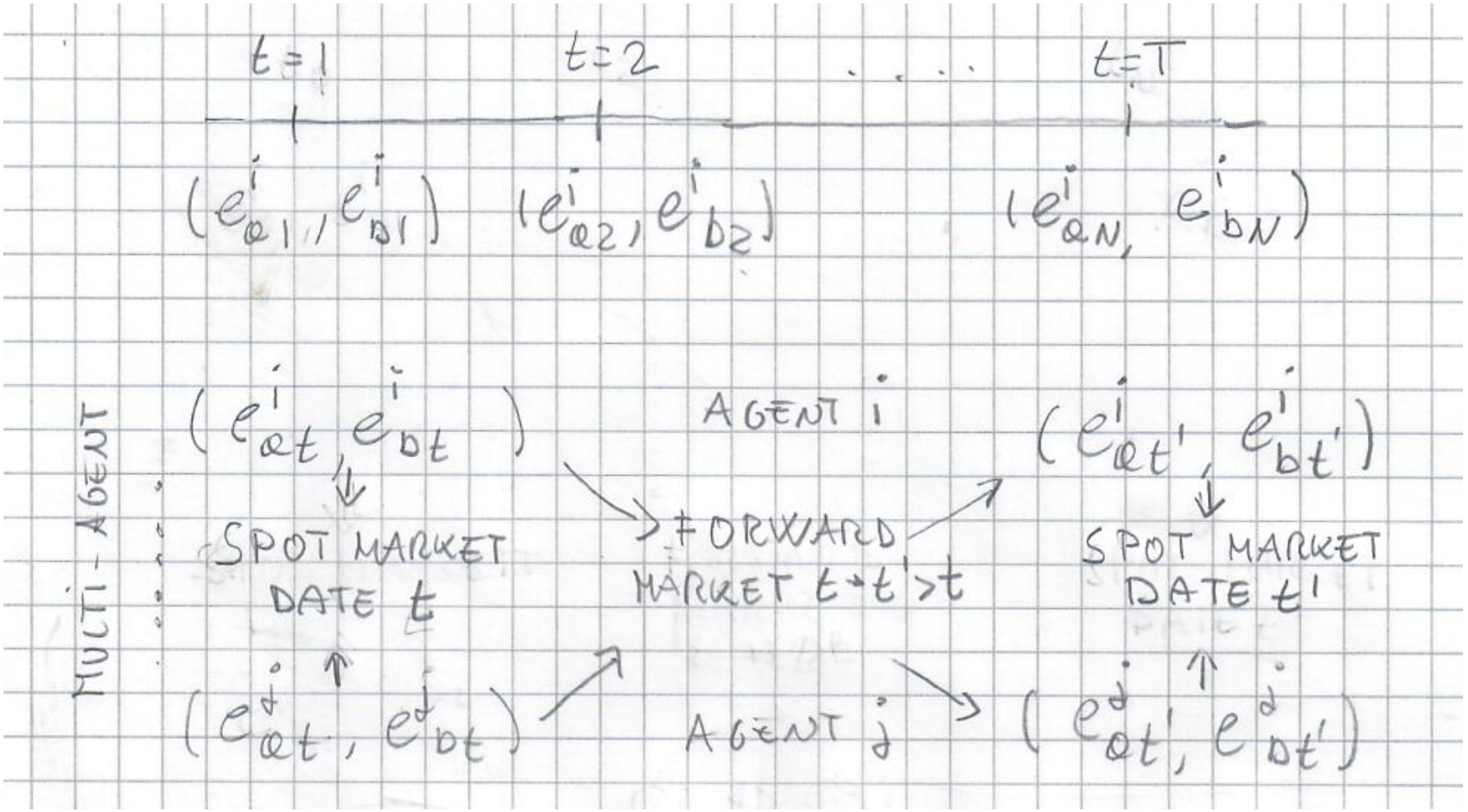
$$p_a c_a^i + p_b c_b^i \leq p_a e_a^i + p_b e_b^i \ .$$

LOP

Why, with N good, barter with $\sim N^2$ d.o.f. is equivalent to the price system with $\sim N$ d.o.f? Hidden constraints (apple, bananas and coconut in the picture):



Multidate


t=1
t=2
. . . .
t=T
(e^i_{a1}, e^i_{b1})
(e^i_{a2}, e^i_{b2})
(e^i_{aN}, e^i_{bN})
MULTI-AGENT
(e^i_{at}, e^i_{bt})
AGENT i
$(e^i_{at'}, e^i_{bt'})$
SPOT MARKET
DATE t
FORWARD
MARKET t→t'>t
SPOT MARKET
DATE t'
(e^j_{at}, e^j_{bt})
AGENT j
$(e^j_{at'}, e^j_{bt'})$

Multidate prices

SPOT (t)

$S_t \leftrightarrow \mathbb{E}_t$

P_{qt}

$D_t \leftrightarrow \mathbb{E}_t$

P_{bt}

FORWARD ($t \to t'$)

$S_{t'} \leftrightarrow \mathbb{E}_t$

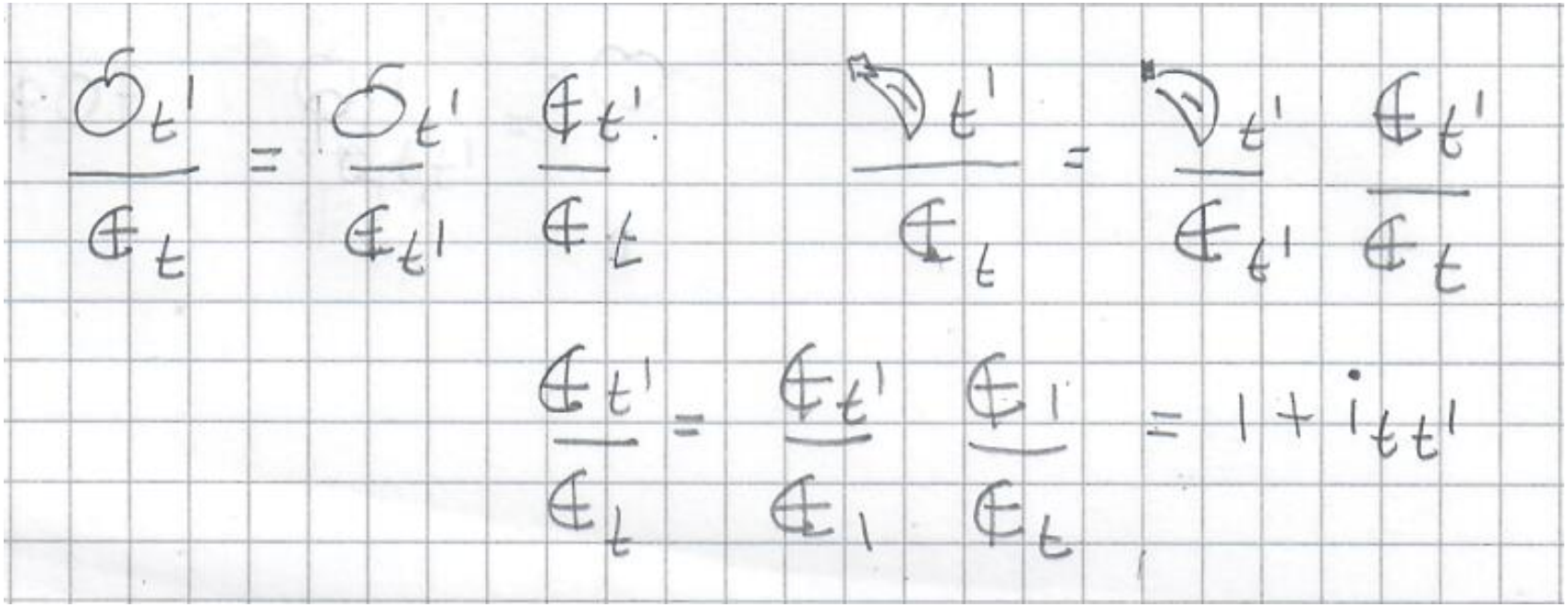
$P_{qtt'}$

$D_{t'} \leftrightarrow \mathbb{E}_t$

$P_{btt'}$

$N \times T$ # PRICES $N \times T \frac{(T-1)}{2}$

Multidate LOP



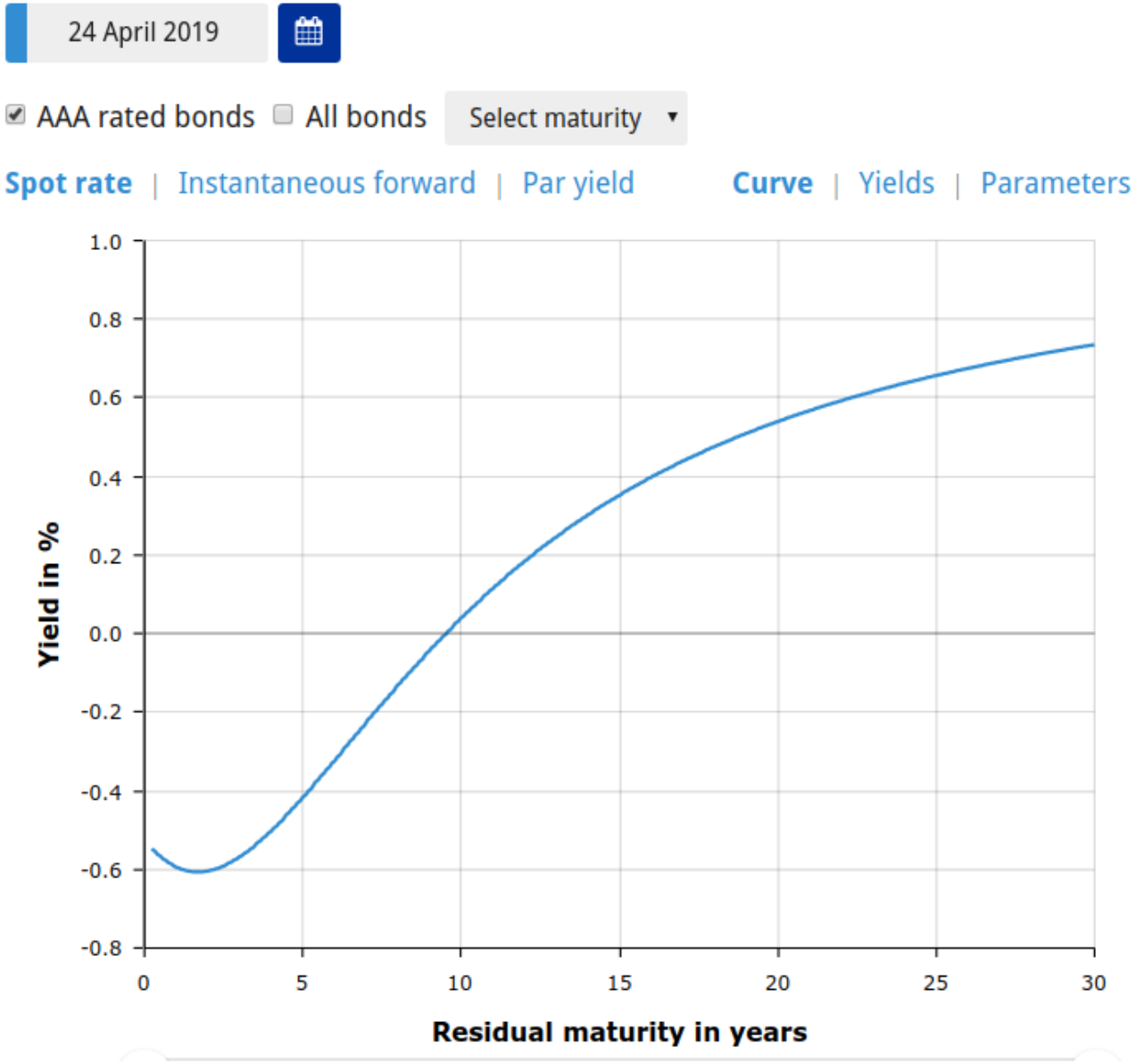
$1 + i_{tt'}$ is the gross forward nominal interest rate

$$p_{att'} = p_{at'}\left(1 + i_{tt'}\right) \quad p_{btt'} = p_{bt'}\left(1 + i_{tt'}\right) \quad 1 + i_{tt'} = \frac{1 + i_{t'}}{1 + i_t}$$

where $i_t = i_{1t}$ is the nominal interest rate. <u>The problem factorize</u>: forward prices are obtained from spot prices and interest rates.

Yield curve

Euro area yield curves (annualized rates) from ECB.



Multidate LOP again

$$\underbrace{\boxed{\frac{€_{t'}}{€_t}}}_{1+i} = \frac{€_{t'}}{\$_{t'}}\,\frac{\$_{t'}}{\$_t}\,\frac{\$_t}{€_t} = \underbrace{\boxed{\frac{\$_{t'}}{\$_t}}}_{1+r_a}\,\underbrace{\boxed{\frac{\$_t}{€_t}\,\frac{€_{t'}}{\$_{t'}}}}_{1+\pi_a}$$

$1 + r_a$ is the gross real interest rate on apples; $1 + \pi_a$ is the inflation rate of apples and one recovers the Fisher's equation

$$1 + i = (1 + r_a)(1 + \pi_a) = (1 + r_b)(1 + \pi_b) = \dots$$

The nominal interest rate is unique, but the real interest rates are legions.

microeconomics + time = financial economics

Financial Economics is microeconomics with

- a homogeneous consumption good (the general problem factorizes!);
- Walrasian, centralized and instantaneous price fixing;
- symmetric information;
- more interesting with uncertainty;
- possibly incomplete markets.

Payoffs

Consider a simple model with two dates:

- t_0: first date (today), J securities are traded and prices fixed,
- t_1: second date (tomorrow), a state of the world s among the S possible states is realized and each security j pays a payoff $x_{j,s}$.

Definition

Payoff Matrix ($J \times S$):

$$X = \left\| \begin{matrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_J \end{matrix} \right\| = \left\| \begin{matrix} x_{1,1} & \cdots & x_{1,S} \\ \vdots & \cdots & \vdots \\ x_{J,1} & \cdots & x_{J,S} \end{matrix} \right\| .$$

$x_{i,s}$: payoff of security i if state s is realized,

$\mathbf{x}_i$: row vector of payoffs of asset i,

$\mathbf{x}_{\cdot,s}$: column vector of payoffs in state s.

Portfolios

Definition

Portfolio: $\mathbf{h} = (h_1 \ , \ \ldots \ , \ h_J)$ is an holding of the J securities.

The payoff of the portfolio in state s is $\sum_j^J h_j x_{j,s}$. Since $\mathbf{h}$ represents a row vector, the payoff is a row vector

$$\mathbf{h}X = \left(\sum_j^J h_j x_{j,1} \ , \ \ldots \ , \ \sum_j^J h_j x_{j,S} \right) \ .$$

Position in asset j is long if $h_j < 0$ and short if $h_j < 0$ (sometimes we can exclude short positions).

It is generally desirable to have a market that allows to build portfolios with diversified payoffs.

Asset Span

Definition

Asset Span: assume no constraints on portfolios, it is the set of all possible payoff column vectors

$$\mathcal{M} = \{\mathbf{z} \in \mathbb{R}^S \mid \exists \mathbf{h} \in \mathbb{R}^J \; s.t. \; \mathbf{z} = \mathbf{h}X\}.$$

- If $\mathcal{M} = \mathbb{R}^S$, the market is *complete* (all payoff structures can be achieved),
- if $\mathcal{M} \subset \mathbb{R}^S$, the market is *incomplete*,
- a security j is *redundant* if $\exists \mathbf{h}, h_j = 0$ such that $\mathbf{x}_j = \mathbf{h}X$, i.e. if its payoff structure can be replicated by a portfolio that does not contain it.

Theorem

The market is complete iif Rank$\{X\} = S$. *There are not redundant securities iif Rank*$\{X\} = J$.

Prices and Returns

Prices of securities at time 0 are $\mathbf{p} = (p_1, \ldots, p_J)^t$ and *gross returns* are defined as

$$\mathbf{r}_j = \frac{\mathbf{x}_j}{p_j} = \left(\frac{x_{j,1}}{p_j}, \ldots, \frac{x_{j,S}}{p_j} \right)^t .$$

Net returns are $\mathbf{r}_j - \mathbf{1}$.

In what follows we shall make use of the notion of *security market* as the couple $\{X, \mathbf{p}\}$: a given set of securities and their prices. In this analysis the prices $\mathbf{p} = (p_1, \ldots, p_J)$ are considered as given.

Law of One Price

Definition

The security market $\{X, \mathbf{p}\}$ satisfies the *Law of One Price* (LOP) if $\forall \mathbf{h}, \mathbf{h}'$ s.t. $\mathbf{h}X = \mathbf{h}'X$ it is $\mathbf{hp} = \mathbf{h}'\mathbf{p}$.

- Portfolios with the same payoffs have the same price.

Theorem

The Law of One Price (LOP) holds iff any portfolio with zero payoff has zero price, i.e. $\forall \mathbf{h}$ *s.t.* $\mathbf{h}X = 0$ *it is* $\mathbf{hp} = 0$.

Assume that the LOP does not hold, then:

- $\exists \mathbf{h}_0$ such that $\mathbf{h}_0 X = 0$ and $\mathbf{h}_0\mathbf{p} = c > 0$,
- but notice that $\lambda \mathbf{h}_0 X = 0$ and $(\lambda \mathbf{h}_0)\mathbf{p} = \lambda(\mathbf{h}_0\mathbf{p}) = \lambda c$ so the zero payoff portfolio can be purchased at any price,
- take a portfolio $\mathbf{h}$ s.t. $\mathbf{h}X = \mathbf{z}$ and $\mathbf{hp} = c'$ then $\mathbf{h} + \lambda \mathbf{h}_0$ has the same payoff and price $c' + \lambda c$. Thus *any payoff can be purchased at any price*.

Payoff Pricing

Definition

Given a securities market $\{X, \mathbf{p}\}$, define the *payoff pricing* correspondence

$$\begin{aligned} q(\cdot) : \ \mathcal{M} &\to \mathbb{R} \\ \mathbf{z} &\to q(\mathbf{z}) = \{w = \mathbf{hp} \,|\, \text{s.t. } \mathbf{z} = \mathbf{h}X\} \,. \end{aligned}$$

There could be more than one portfolio $\mathbf{h}$ such that $\mathbf{z} = \mathbf{h}X$ and more than one w.

Theorem

The LOP holds iff $q(\cdot)$ is a functional on $\mathcal{M}$.

Remember: a functional is a linear map. When $q(\cdot)$ is a functional, it is called the *payoff pricing functional*.
When the L.O.P. does not hold it is $q(\mathbf{z}) = \mathbb{R}$ for any $\mathbf{z} \in \mathcal{M}$.

State Claims

Definition

Let e_s be the *state claim* or *Arrow security* associated to state s, that is, the payoff of e_s is 1 if state s is realized and zero otherwise:

$$\mathbf{e}_s = (0, \ldots, 1, \ldots, 0)$$

where the 1 is in the s-th place.

- If the market is complete, then $\forall s$, $\mathbf{e}_s \in \mathcal{M}$ and one can price these states using $q(\cdot)$: *state price* $\rightarrow q_s = q(\mathbf{e}_s)$,
- If the L.O.P. is valid, the linear structure of $q(\cdot)$ gives $q(\mathbf{z}) = \sum_s z_s q_s = \mathbf{z}\mathbf{q}$, with $\mathbf{q} = (q_1, \ldots, q_S)^t$,
- consider the security j with payoff $\mathbf{x}_j$. It is $q(\mathbf{x}_j) = q\left(\sum_s x_{j,s} z_s\right) = \sum_s x_{j,s} q_s$ so that one has $p_j = \mathbf{x}_j \mathbf{q}$ and, in matrix notation, $\mathbf{p} = X\mathbf{q}$.

Pricing Functional

The functional $\mathbf{q}$, defined on $\mathcal{M}$, can be extended to the entire $\mathbb{R}^S$. Let $(\mathbf{z}_1, \ldots, \mathbf{z}_K)$ be a basis for $\mathcal{M}$ and complete it $(\mathbf{z}_1, \ldots, \mathbf{z}_K, \mathbf{z}_{K+1}, \ldots, \mathbf{z}_S)$ and set

$$q^*(\mathbf{z}_j) = \begin{cases} q(\mathbf{z}_j) & j = 1 \ldots K \\ p_j^* & j = K+1 \ldots S \end{cases}$$

- In this way one obtains a *pricing model*.

Definition

A *valuation functional* is a linear map

$$\begin{aligned} Q : &\mathbb{R}^S \to \mathbb{R} \\ &\mathbf{z} \to Q(\mathbf{z}) \end{aligned}$$

such that $\mathbf{Q}/\mathcal{M} = \mathbf{q}$, that is, the restriction to the asset span is equal to the payoff pricing functional.

Preliminary

Definition

If $\mathbf{x} \in \mathbb{R}^n$ is a vector, we say

- $\mathbf{x} \geq 0$ if $x_j \geq 0$, $\forall j \in 1 \ldots n$;
- $\mathbf{x} > 0$ if $x_j \geq 0$ $\forall j \in 1 \ldots n$; $\exists k$ s.t. $x_k > 0$,
- $\mathbf{x} >> 0$ if $x_j > 0$, $\forall j \in 1 \ldots n$.

Definition of Arbitrage

Consider a security market $\{X, \mathbf{p}\}$, then

Definition

A portfolio $\mathbf{h}$ is a *strong arbitrage* if $\mathbf{h}X \geq 0$ and $\mathbf{hp} < 0$. A portfolio $\mathbf{h}$ is a *weak arbitrage* if $\mathbf{h}X > 0$ and $\mathbf{hp} = 0$.

- A strong arbitrage corresponds to a *risk-less reward at time zero*, while a weak arbitrage corresponds to a *costless possible reward at time one*,
- if a market has at least a portfolio which is a strong arbitrage or weak arbitrage, it is said to admit strong or weak arbitrage. A market admits arbitrage if it has at least one weak or one strong arbitrage portfolio.

Positive Payoff Pricing Functional and Arbitrage

- Let us negate the strong arbitrage condition: $!(\exists \mathbf{h},\ \mathbf{h}X \geq 0 \text{ and } \mathbf{hp} < 0) \rightarrow \forall \mathbf{h}$, if $\mathbf{h}X \geq 0$ it is $\mathbf{hp} \geq 0$. This means that for a $\mathbf{z} \in \mathcal{M}$, if it is $\mathbf{z} \geq 0$ then $q(\mathbf{z}) \geq 0$,
- the negation of the weak arbitrage condition: $!(\exists \mathbf{h},\ \mathbf{h}X > 0 \text{ and } \mathbf{hp} = 0) \rightarrow \forall \mathbf{h}$, if $\mathbf{h}X > 0$ it is $\mathbf{hp} > 0$. The negation implies that if $\mathbf{z} \in \mathcal{M}$, $\mathbf{z} > 0$ then $q(\mathbf{z}) > 0$.

Definition

the payoff pricing functional $\mathbf{q}$ is *positive* if $\forall \mathbf{z} \geq 0$, it is $q(\mathbf{z}) \geq 0$ and it is *strictly positive* if $\forall \mathbf{z} > 0$, it is $q(\mathbf{z}) > 0$.

- If $\mathbf{q}$ is positive, there is no strong arbitrage,
- if $\mathbf{q}$ is strictly positive, then there is no weak arbitrage, but since $q(\mathbf{z} = 0) = 0$ there is no strong arbitrage too. Thus, there is no arbitrage.

Farka's Lemma

Theorem

In a security market $\{X, \mathbf{p}\}$

- *or there exists a* $\mathbf{q} \in \mathbb{R}^s$, $\mathbf{q} \geq 0$ *such that* $\mathbf{p} = X\mathbf{q}$,
- *or there exists a* $\mathbf{h} \in \mathbb{R}^J$ *such that* $\mathbf{h}X \geq 0$ *and* $\mathbf{hp} < 0$.

No strong arbitrage iff there exists a $\mathbf{q}$ that satisfies the first statement. If $\mathbf{z} = \mathbf{h}X \in \mathcal{M}$, then $q(\mathbf{z}) = \mathbf{h}\,\mathbf{p} = \mathbf{h}X\mathbf{q} = \mathbf{z}\,\mathbf{q}$.

The price vector $\mathbf{p} \in C$, the *convex cone* of the column vector of $X = ||\mathbf{x}_{\cdot,1} \ldots \mathbf{x}_{\cdot,S}||$

$$C = \left\{ \mathbf{h} \in \mathbb{R}^J \middle| \mathbf{h} = \sum_{s=1}^{S} \lambda_s \mathbf{x}_{\cdot,s},\ \lambda_s \geq 0 \right\} .$$

$\mathbf{q}$ define a positive valuation functional: $Q(\mathbf{z}) = \mathbf{z}\,\mathbf{q}$ for $\mathbf{z} \in \mathbb{R}^s$.

Stiemke's Lemma

Theorem

In a security market $\{X, \mathbf{p}\}$

- *or there exists a* $\mathbf{q} \in \mathbb{R}^s$, $\mathbf{q} >> 0$ *such that* $\mathbf{p} = X\mathbf{q}$,
- *or there exists a* $\mathbf{h} \in \mathbb{R}^J$ *such that* $\mathbf{h}X \geq 0$ *and* $\mathbf{hp} < 0$ *or* $\mathbf{h}X > 0$ *and* $\mathbf{hp} \leq 0$.

No arbitrage iff there exists a $\mathbf{q}$ that satisfies the first statement.

The price vector $\mathbf{p} \in C^0$, the *relative interior of convex cone* of the column vector of $X = ||\mathbf{x}_{\cdot,1} \ldots \mathbf{x}_{\cdot,S}||$

$$C = \left\{ \mathbf{h} \in \mathbb{R}^J \middle| \mathbf{h} = \sum_{s=1}^{S} \lambda_s \mathbf{x}_{\cdot,s},\ \lambda_s > 0 \right\} .$$

$\mathbf{q}$ define a strictly positive valuation functional: $Q(\mathbf{z}) = \mathbf{z}\,\mathbf{q}$ for $\mathbf{z} \in \mathbb{R}^s$.

The Fundamental Theorem of Finance

Theorem (Fundamental Theorem of Finance)

- *Security prices exclude strong arbitrage iff there exists a positive valuation functional.*
- *Security prices exclude arbitrage iff there exists a strictly positive valuation functional.*

The discussion and theorem above are about the existence of a valuation functional, not its consequences. Notice that $Q(\mathbf{e}_s) = q_s$, so two different vectors $\mathbf{q}$ define two different valuation functionals. In practice the *positive valuation functional* is the $\mathbf{q}$ which solves

$$X\mathbf{q} = \mathbf{p} \quad \mathbf{q} \geq 0\,,$$

while the *strictly positive valuation functional* solves

$$X\mathbf{q} = \mathbf{p} \quad \mathbf{q} >> 0\,.$$

Uniqueness of Valuation Functional

In general the number of solution of $X\mathbf{q} = \mathbf{p}$ depends on X. It is $X : \mathbb{R}^S \to \mathbb{R}^J$ then

- if $\text{Rank}(X) \geq J$, since $\text{Im}(X) = \mathbb{R}^J$ there exists at least one solution for any $\mathbf{p} \in \mathbb{R}^J$
- if $\text{Rank}(X) = S$ then there exists exactly one solution for any $\mathbf{p} \in \mathbb{R}^J$.

These statements do not say anything about the positivity or strict positivity of the solution.

Arbitrage-Free Bounds

The absence of strong arbitrage (or arbitrage) does imply the existence of positive (or strictly positive) state prices. Since these prices are not unique, the same payoff structures or *contingent claims* $\mathbf{z} \in \mathbb{R}^S$ could be priced differently.

We can however derive bounds.

Definition

Let $\{X, \mathbf{p}\}$ be without strong arbitrage (or arbitrage), then for any $\mathbf{z} \in \mathbb{R}^S$ define the *upper and lower bounds*

$$Q_u(\mathbf{z}) = \sup_{\mathbf{q} \geq 0 \, (\mathbf{q} >> 0)} \{\mathbf{q}\mathbf{z} \, : \, \mathbf{p} = X\mathbf{q}\}$$

$$Q_l(\mathbf{z}) = \inf_{\mathbf{q} \geq 0 \, (\mathbf{q} >> 0)} \{\mathbf{q}\mathbf{z} \, : \, \mathbf{p} = X\mathbf{q}\}$$

$Q_u(\mathbf{z})$ and $Q_l(\mathbf{z})$ represent the extremal prices of a payoff consistent with the lack of arbitrage.

Riskless Security

Consider the riskless security $(c, \ldots, c)$ that pays c in every state of the world. If $\mathbf{c} \in \mathcal{M}$, then $(\lambda c, \ldots, \lambda c) \in \mathcal{M}$ and if the L.O.P. applies, $q(\lambda \mathbf{c}) = \lambda q(\mathbf{c})$. This implies that

Corollary

If the L.O.P. is valid, the return of every riskless security $\mathbf{z}$ *is the same:* $R_f = \mathbf{z}/q(\mathbf{z})$.

In terms of the state prices (if they exist) it is

$$R_f = \frac{1}{\sum_s q_s}.$$

Notice that if the valuation functional is not positive, then it could be $R_f < 0$ or R_f is not defined because $\sum_s q_s = 0$.

Risk Neutral Probabilities I

Definition

Assume that a positive valuation functional exists, one can define the *risk neutral probabilities*

$$\pi_s = \frac{q_s}{\sum_s q_s}\,.$$

Notice that $\sum_s \pi_s = 1$, $\pi_s \geq 0$.

Risk neutral probabilities are not probabilities.

Risk Neutral Probabilities II

Previous results can be restated using the $\boldsymbol{\pi} = (\pi_1, \ldots, \pi_S)$:

- value of a contingent claim $\mathbf{z}$
  $Q(\mathbf{z}) = \mathbf{q}\mathbf{z} = \sum_s q_s z_s = {R_f}^{-1} \sum_s \pi_s z_s = {R_f}^{-1} \mathrm{E}_{\boldsymbol{\pi}}[\mathbf{z}]$ where $\mathrm{E}_{\boldsymbol{\pi}}[\cdot]$ is the expected value relative to the probability measure $\boldsymbol{\pi}$,
- price of security $\mathbf{x}_j$: $p_j = \mathbf{x}\mathbf{q}_j = {R_f}^{-1} \mathrm{E}_{\boldsymbol{\pi}}[\mathbf{x}_j]$,
- the valuation functional $Q(\mathbf{z}) = {R_f}^{-1} \mathrm{E}_{\boldsymbol{\pi}}[\mathbf{z}]$,
- the relationship of the $\boldsymbol{\pi}$'s with arbitrage is the same of the $\mathbf{q}$'s:

  no strong arbitrage $\leftrightarrow$ the $\boldsymbol{\pi}$'s exists
  no arbitrage $\leftrightarrow$ the $\boldsymbol{\pi}$'s are strictly positive